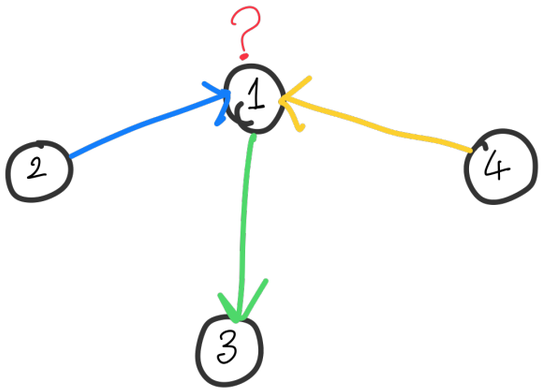
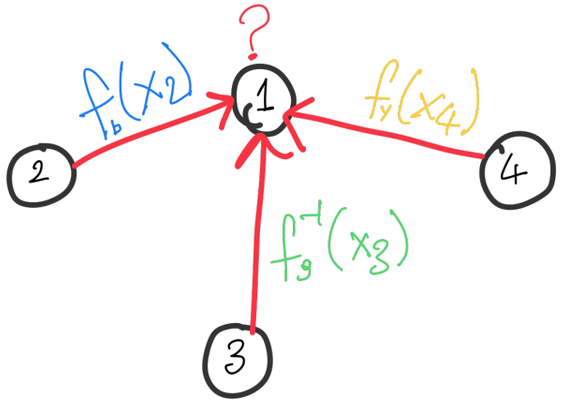
Multi-relational Propagation (MRP)

from Message-Passing View Point



Relation Types
Blue
Green
Yellow

Assoc. message functions
fb()
fg()
fy()



Multi-relational Directed Graph

$\mathcal{G}(\mathcal{V}, \mathcal{E}, \mathcal{P})$

Relational Local Generative Model

$$x_i = \eta_{\mathrm{p}} x_j + \tau_{\mathrm{p}} + \epsilon$$
$$\forall \mathrm{r}(i,j) = \mathrm{p}$$
$$\epsilon \sim \mathcal{N}(0, 1/\omega_{\mathrm{p}})$$

Relational Local Estimate

$$\underset{x_i}{\operatorname{argmin}} \sum_{\mathrm{p} \in \mathcal{P} \cup \mathcal{P}^{-1}} \sum_{\mathrm{r}(i,j)=\mathrm{p}} \omega_{\mathrm{p}} (x_i - \eta_{\mathrm{p}} x_j - \tau_{\mathrm{p}})^2$$

Neighborhood Aggregation

$$\frac{\sum_{\mathrm{p} \in \mathcal{P} \cup \mathcal{P}^{-1}} \sum_{\mathrm{r}(i,j)=\mathrm{p}} \omega_{\mathrm{p}} f_{\mathrm{p}}(x_j)}{\sum_{\mathrm{p} \in \mathcal{P} \cup \mathcal{P}^{-1}} \sum_{\mathrm{r}(i,j)=\mathrm{p}} \omega_{\mathrm{p}}}$$

A relation type $\mathrm{p} \in \mathcal{P}$

$i \xleftarrow{\mathrm{p}} j$ $\quad \mathrm{r}(i,j) = \mathrm{p} \quad \mathrm{r}(j,i) = \mathrm{p}^{-1}$

Relational Transformation

$$f_{\mathrm{p}}(x) = \eta_{\mathrm{p}} x + \tau_{\mathrm{p}}$$